01 · LESSON

Natural selection and evolution

9A Genetics and evolution · Lesson 7 of 7

Date _____ Class _____

Today I can

- ☐ I can explain how natural selection works on inherited variation when the environment changes, using the peppered moth and the Salton Sea tilapia.
- ☐ I can describe the theory of evolution by natural selection and explain how natural selection can produce a new species.

What happens today

1. Do Now: pale or dark?
2. Learn: Natural selection: the peppered moth · Salty seas: the Salton Sea tilapia · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 18 to 20 · Workbook pages 13 to 14

What I learned today (one or two sentences)

02 · KEY VOCABULARY

Key Vocabulary

Term	Definition
natural selection	organisms better adapted to their environment are more likely to survive, reproduce and pass on their genes
evolution	a change over time in the characteristics of organisms, sometimes making a new species
inherited variation	differences between individuals caused by genes, passed on to offspring
gene	a section of DNA that carries the instructions for one characteristic
reproduce	to produce offspring
generation	all the offspring born at about the same time; parents, children and grandchildren are three generations
hypothesis	an idea, based on observations, that can be tested
theory	an explanation that has been tested many times and is accepted by scientists
species	a group of organisms that can breed together to produce offspring that can also breed
camouflage	colours or patterns that make an organism hard to see
cloning	making a new organism with exactly the same genetic information as another

03 · QUICK REVIEW

Quick Review

pale or dark?

Peppered moths show inherited variation: most are pale and speckled, but some have genes that make them black.

1 What causes inherited variation?

2 Which form would a bird spot more easily on a pale tree? On a tree blackened by soot?

3 What is a species?

04 · CORE

Core 1 of 3

Everyone does these. Answer on the card or under it.

1 State what causes inherited variation.

2 Many times a year, a certain river suddenly floods. Over many years, a species of lizard living by the river evolved longer legs. Darwin might have explained this using the sentences in the table. Number them 1 to 5 in the correct order.

Order	Sentence
	In the next generation, more of the lizards have longer legs.
	The surviving lizards produce offspring that also have longer legs.
	The lizards with slightly longer legs are more likely to survive the flood.
	This process of natural selection repeats over and over again.
	When the river floods, the lizards with slightly longer legs run faster.

Core 2 of 3

Everyone does these. Answer on the card or under it.

3 In the 1850s most peppered moths around Manchester were pale. By 1895, 98% were black. Factories built in those years covered the trees in soot.

a) Explain why black moths survived better than pale moths around Manchester at the end of the 19th century.

Core 3 of 3

Everyone does these. Answer on the card or under it.

3 *continued*

b) Manchester now has many fewer factories and the trees are no longer black. Predict how the number of black moths has changed, and use natural selection to explain your prediction.

Challenge 1 of 3

When Core is finished. Full sentences.

C1 Explain how natural selection can produce a new species if two populations of the same organism become separated, for example on two islands. (Hint: think about the definition of a species.)

Challenge 2 of 3

When Core is finished. Full sentences.

C2 Woolly mammoths became extinct about 4000 years ago. Mammoths and elephants evolved from one species about 6 million years ago. Mammoths had 58 chromosomes.

a) How many chromosomes would a mammoth sperm cell have?

b) Suggest how mammoths may have evolved thick hair, when the species they evolved from had very little hair.

Challenge 3 of 3

When Core is finished. Full sentences.

C2 *continued*

c) Explain why scientists could clone a mammoth from frozen remains but could not clone a dinosaur.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 What is natural selection?

2 What is evolution?

3 Explain why dark peppered moths became more common near Manchester in the 1800s.

Today I can: Goal 1 ☐ Goal 2 ☐